Project Proposal

Start Assignment

- Due Friday by 5pm
- Points 10
- Submitting a file upload
- File Types pdf

Groups

This is a group project for groups of **2-3 students**.

Project Topic

This is your chance to choose a project of your choice: from identifying a problem to designing and implementing a solution to evaluating that solution. You have

Project Scope

Your major effort this semester outside of reading papers for class is to do a research project. There are three styles of research project:

- **Full project:** Measure/Design/Evaluate: identify an opportunity, measure the potential benefit of addressing the opportunity, design and implement a solution, evaluate the solution
- **Exploratory project**: Measure only: identify several opportunities for research by measuring the potential benefits of a new design. For this, you identify situations where an application/system performs poorly (latency, throughput, efficiency, predictability, power consumption) and demonstrate there is potential for a better solution.
- **Reproduce/evaluate:** Reproduce the design of another paper (using their code or a reimplementation) and evaluate it in a broader set of situations, such as more hardware configurations, workload configurations (inputs, # of threads, etc.) or workloads.

Project ideas

Here are a few project ideas to start your thinking, but you are welcome and encouraged to propose your own that interests your group as long as its related to operating systems. If you select one of the following ideas, you will need to develop it into a full project proposal.

- **Optimize a Linux policy:** choose a Linux policy (**https://docs.google.com/spreadsheets/d/1mz9Xbv5EIgYMgVP8jMDsfLRSA9mB7UIuGtEok5usp=sharing** **(https://docs.google.com/spreadsheets/d/1mz9Xbv5EIgYMgVP8jMDsfLRSA9mB7UIuGtEok5D7Sg8/edusp=sharing)** , **lwn.net,** **(https://lwn.net/)** **Linux Plumbers Conference** **(https://lpc.events/event/18/)**), look at how changing the policy affects performance. Find a mix

workloads that work better with different policy choices and characterize the policy's performance before and after the change.

- **Make a Linux policy extensible:** Find a Linux policy that is not cleanly separated from mechanism, and redesign the code to allow the policy to be extensible, such as controllable from userspace, and show that changing the policy has some benefit to some application
- **Optimize a file system**: It is possible to design a better file systems for specific workloads, such as applications with millions of small files, AI KV-caches, to be under distributed append-only file systems, or to work better with a FTL. Pick and application and design a better file system for it (I recommend using **FUSE** (https://www.kernel.org/doc/html/next/filesystems/fuse.html) for this but you're also welcome to write it directly in the kernel).
- **Memory:** Memory is a scarce and expensive resource today there are ways to optimize its use but they have performance overheads. Characterize and develop policies to improve trade-offs such as huge pages (huge pages improve the performance of TLBs but also increase memory fragmentation) or page allocation (page allocation performance bottlenecks of moving memory to "farther" memory tiers such as DDR4, **CXL** (https://dl.acm.org/doi/full/10.1145/3669900) , and flash).

AI use in this project

For the project, it is up to you on whether and how to use AI for coding. Before submitting the project, your group will need to discuss and agree to your own group's AI policy. If you decide to use AI, you will be evaluated based on whether you effectively used AI and you will need to document how you used AI including: example prompts, what parts of the code were AI generated, how your AI use aided your learning, if and when the AI did not produce what you were expecting. If you decide not to use AI, you will need to document this decision and discuss how it the decision impacts your groups' learning in the course.

Since this course is developing research skills including creating research and communicating them, you **may not** use AI for idea generation, writing, or creating presentations. I know that these pieces are hard, but unfortunately they will not get easier without practice nor help you develop tools for understanding when ideas are worth pursuing or when presentation or writing are well done.

What to produce

For the project proposal, your team should determine exactly what problem you will solve. In your project proposal, you should clearly state:

- the problem you are planning to solve
- the motivation for why this problem is important
- your initial plans for approaching the problem
- your proposed methodology for evaluation.
- what resources you will use / need for the project, such as hardware or software tools.
- your plan for whether and how you are using AI in the project

This report should contain references to at least one related background paper that you have read. Your write-up should be approximately a full page in length (or 500-1000 words). Like for mini project 3, submit a pdf and share your overleaf/google docs file with my email: sjmcallister@wisc.edu.

Preview of next assignment:

An important component in research is placing your contribution in the context of past work, which involves differentiating your work from previous work. (There is nothing worse than working on a problem, only to discover several months into the work, that someone has already solved it!) To ensure that you have found the applicable background material before beginning your own work, I would like to see a summary of the relevant papers that you have read (or, alas, skimmed). While the number of relevant papers will vary depending upon your project, you should search for 8 to 10 references and write 500-1000 words. To find relevant papers, you may want to check out the resources at the end of this page. Your report should contain the appropriate citations in a Reference section (I would suggest using bibtex for this). This write-up may later become the Related Work section of your final report (of course, you may need to add references, especially if the direction of your contribution changes).

In addition, if you received any follow-up questions on your project proposal, please address them in your related work report.